

# Bridging Design and Quality: A Technical Overview of IFMs in Manufacturing

Chih-Fan Hsu<sup>1</sup>, Yung-Huei Chiou<sup>1</sup>, Po-Han Huang<sup>1</sup>, Yu-chi Liao<sup>1</sup>,  
Shu Huei Yang<sup>1</sup>, Jyun-Ze Tang<sup>1</sup>, Yu-Lun Chang<sup>1</sup>, and Ming-Ching Chang<sup>2</sup>

<sup>1</sup>Inventec Corporation

<sup>2</sup>University at Albany, State University of New York

{hsuchihfan, chiouyh25, alan880603}@gmail.com, yukiandyoki@moegi.waseda.jp

{cindy381020, col360785, helenafterglow}@gmail.com, mchang2@albany.edu

**Abstract**—Industrial Foundation Models (IFMs) represent a transformative shift in smart manufacturing, offering unprecedented multi-task generalization through multimodal data integration. However, the transition from general-purpose AI to factory-floor deployment is hindered by stringent requirements for precision and the lack of standardized benchmarks. While existing surveys focus on high-level applications, this paper provides a granular technical roadmap centered on the data-centric and metric-centric challenges of the field. We deliver an exhaustive analysis of publicly available datasets and expose the critical misalignment between standard machine learning scores and practical manufacturing constraints. By consolidating these technical pillars, this work equips researchers with the essential resources to develop trustworthy, production-ready IFMs for the next generation of industrial intelligence.

**Index Terms**—Industrial Foundation Model, Manufacturing

## I. INTRODUCTION

The rapid development of industrial foundation models (IFMs) is reshaping the landscape of smart manufacturing. Unlike traditional task-specific machine learning (ML) systems, IFMs leverage large-scale multimodal pretraining to capture cross-domain knowledge and generalize across diverse tasks (Figure 1). This paradigm shift promises not only improvements in efficiency and adaptability but also a unified framework for integrating heterogeneous data sources, such as sensor streams, visual inspections, computer-aided design models, and industrial logs.

Despite their potential, the deployment of IFMs in manufacturing faces significant hurdles. Industrial environments demand a higher degree of precision, reliability, and interpretability than general-purpose applications. Furthermore, the lack of high-quality, domain-specific datasets and the absence of standardized evaluation metrics have led to reproducibility issues, making it difficult to benchmark different models objectively. While general foundation models like GPT-5 or CLIP provide a baseline, adapting them to handle complex industrial geometries or rare anomaly detection requires specialized technical frameworks.

This paper provides a comprehensive technical overview of IFMs, with a specific focus on bridging the gap between IFMs and practical manufacturing requirements. While existing surveys primarily emphasize Large Language Models or general training strategies, we pivot the focus toward including Vision

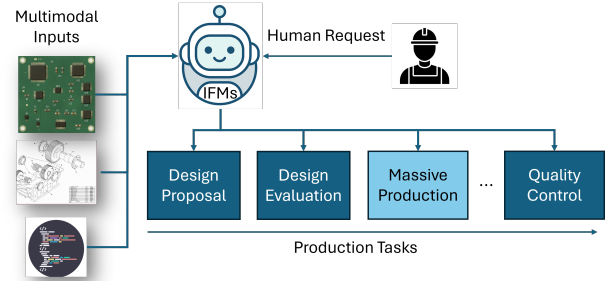


Fig. 1: Industrial Foundation Models (IFMs) take multi-modal inputs to handle various tasks in manufacturing. The light blue sector is excluded from this overview because of the scope.

Foundation Models (VFM), publicly available datasets, and standardized evaluation protocols for trustworthy deployment.

The main contributions of this overview are as follows:

- We provide a comprehensive review of publicly available datasets that support design and quality control, highlighting the role of synthetic data in addressing industrial data scarcity.
- We analyze current evaluation metrics and identify the misalignment between standard ML scores and practical manufacturing objectives (e.g., manufacturability and physical constraints).
- We identify existing research gaps and propose future directions, including edge deployment and model explainability.

### A. Related Work

Several recent studies have explored the emergence of foundation models in the industry, yet they differ significantly in scope from this work.

Zhang et al. [1] provided a systematic overview of smart manufacturing challenges and applications, focusing predominantly on the integration of Large-Scale Foundation Models. Similarly, Mustapha [2] conducted a deep dive into how Large Language Models can be trained and fine-tuned for mechanics and product design, emphasizing text-based interactions and framework design.

From a model adaptation perspective, Ren et al. [3] examined IFMs through the lens of system architecture and training

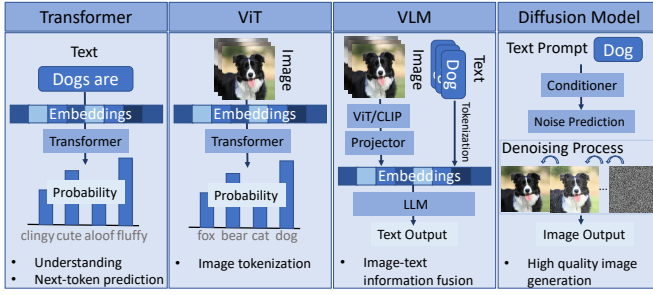


Fig. 2: Recent common model structures enable IFMs.

strategies, discussing how base models can be adapted into domain-specific IFMs. They demonstrated an example named MetaIndex that can handle six tasks, including question answering, scene cognition, and process decision-making. Zhao et al. [4] further categorized developments across data, model, and application levels, offering a high-level view of regional advancements and training techniques.

In contrast to these works, which largely concentrate on LLMs and high-level strategy, our overview provides a more granular technical analysis of the data-centric and metric-centric challenges. We argue that for IFMs to transition from experimental prototypes to production-ready tools, the community needs to prioritize the development of robust datasets and evaluation frameworks that account for the physical and logical rigors of the factory floor.

## II. KEY MODEL STRUCTURE FOR IFMS

Several model structures have emerged as enablers for IFMs, bridging the gap between design, visual inspection, and temporal process understanding (Figure 2).

**Transformer**-based models are neural network architectures utilizing multi-head attention mechanisms and positional encoding. This approach effectively captures dependencies across entire token sequences, addressing the vanishing gradient and information bottleneck issues inherent in Recurrent Neural Networks. Architecturally, a Transformer comprises an encoder and a decoder. The encoder projects input tokens into a latent feature space—effectively embedding temporal and contextual information—while the decoder utilizes this representation alongside query information to auto-regressively generate output sequences. Prior to encoding, a text tokenizer parses raw text into tokens, allowing the encoder to operate on numerical embeddings rather than raw strings. Pre-training on massive corpora equips these models with broad, multi-task capabilities, allowing them to serve as foundation models that can be adapted to diverse downstream applications.

**Vision Transformer (ViT)** adapts this architecture for computer vision by partitioning images into fixed-size patches and treating them as a sequence of embeddings. Similar to text processing, a vanilla ViT utilizes a linear projection layer to map flattened patches into a latent space, acting as an image tokenizer. Unlike language models that rely on next-token prediction, the standard ViT typically prepends a learnable

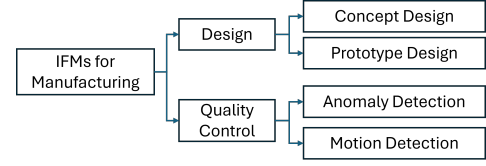


Fig. 3: This overview paper covers two major topic categories in manufacturing, design and quality control.

[CLS] token to the sequence; its state at the final layer serves as the aggregate representation for downstream tasks.

While the ViT fundamentally produces continuous embeddings, these representations can be projected across diverse modalities. In image captioning, for instance, the model maps visual features into a linguistic latent space to generate coherent textual descriptions. The unified token-based design significantly broadens the Transformer’s capability across multiple modalities, most notably in the development of **Vision-Language Models (VLMs)**. VLMs leverage this architectural flexibility by integrating a visual encoder (like ViT) with a language model. In these systems, image patches are treated as visual tokens that reside in the same semantic space as text tokens. This allows the model to perform the tasks that require both visual perception and linguistic understanding, such as Visual Question Answering (VQA), image captioning, or grounded object detection. This versatility is further exemplified by Contrastive Language–Image Pretraining (CLIP), which aligns visual and textual tokens in a shared latent space. Rather than predicting a specific label, CLIP learns to maximize the cosine similarity between the correct image-text pairs while minimizing it for incorrect ones. This advancement has made the research of cross-modal alignment more practical and scalable, forming the foundation for modern cross-modal systems, from text-to-image generators like Stable Diffusion to vision-language models.

**Diffusion Models** build upon these aligned tokens to provide a state-of-the-art framework for generative tasks. By incorporating a text encoder to condition the denoiser during the Backward Process (reverse diffusion), these models synthesize high-fidelity images from pure Gaussian noise based on a text query. This is typically achieved via cross-attention mechanisms where the denoiser attends to the conditioned text embeddings. This synthesis of transformer-based alignment and iterative denoising powers the current frontier of generative AI, enabling tools such as Midjourney and Nano Banana to generate contextually accurate images.

## III. TAXONOMY OF IFM APPLICATIONS IN MANUFACTURING

In this section, we briefly discuss the covered topics in design and quality control categories (Figure 3).

### A. Design Category

Recently, artificial intelligence (AI) approaches have been increasingly applied to support various stages of engineering design, ranging from conceptual development to large-scale

production. Broadly, the process can be viewed in terms of concept design and prototype design. Once the most effective design is determined, automated manufacturing systems can efficiently scale the process to enable massive production.

**Concept design** is an early and creative phase of the engineering design process, where creative exploration and idea generation take precedence. At this stage, there is no single dominant computer-aided tool available to support designers effectively [2]. Instead, the process remains largely intuitive and highly diverse, relying on designers' experience, creativity, and domain knowledge.

In recent years, however, the emergence of generative AI (GAI) has begun to reshape this landscape. GAIs such as pre-trained transformers [5], [6] have been increasingly applied to aid concept design, offering unprecedented capabilities in generating a wide range of novel ideas and alternatives. These models allow for rapid exploration of the design space, helping designers overcome bottlenecks in creativity and iteration.

Traditionally, the evaluation of concept designs has relied heavily on subjective judgment, which can be time-consuming, inconsistent, and limited in scalability. Recently, researchers have explored the use of ML techniques to assess design quality in a more systematic and scalable manner [7], [8]. Such methods enable the automated evaluation of large sets of design candidates within a short time, thereby accelerating the feedback loop and supporting data-driven decision-making in the early stages of engineering design.

For **prototype design**, computer-aided design (CAD) has played a critical role in assisting designers. These tools not only streamline the product development cycle but also enhance precision and reproducibility in design processes. Traditionally, CAD files are generated manually from scratch, which often requires considerable expertise and can be prohibitively time-consuming, particularly for complex geometries or when frequent iterations are needed. Recent advances in GAI have opened new ways for accelerating and automating this workflow. Text-to-CAD generation [9]–[12] enables the automatic translation of natural language descriptions into structured CAD models, lowering the entry barrier for non-experts. Diffusion-based approaches [13] expand this capability by synthesizing CAD models from Gaussian noise, demonstrating the potential to explore novel design spaces beyond human intuition. In addition, image-based generation methods [14] allow the reconstruction of 3D CAD models directly from 2D inputs, bridging the gap between visual perception and manufacturable design. Several methods also parameterize CAD models from hand-drawn sketches [15], [16].

## B. Quality Control Category

Visual inspection remains the bedrock of industrial quality assurance. This section delves into the technologies of anomaly detection (AD) and motion detection. Characterized by the identification of samples deviating from the expected data distribution, AD has emerged as a pivotal component of IFMs for autonomous process monitoring. Unlike traditional

object detection—which is inherently constrained by predefined categories—AD excels in open-set recognition. This allows for the discovery of an exhaustive range of unforeseen irregularities without the prohibitive need for explicit labeling.

**Industrial Visual Anomaly Detection (I-VAD)** mainly focuses on computer vision techniques that automatically identify abnormal regions in images. To avoid collecting a large number of anomalous images, early work focused on unsupervised approaches, categorized into two main types: reconstruction-based methods and feature-based methods. In the former, models are trained on only normal images and anomalies are revealed by residuals between the input and its reconstruction; examples span autoencoder and transformer-style systems and recent diffusion variants, e.g., DRAEM [17], Dinomaly [18], MambaAD [19], AnoDDPM [20], GLAD [21], and DiAD [22]. In the latter, methods build a local feature memory of normal imagery and flag deviations at test time, e.g., ReConPatch [23], PaDiM [24], PatchCore [25], TrustMAE [26], and PNI [27]. While both families typically train (or at least index) on abundant normal data, generalization to novel domains or previously unseen anomalies remains a central deployment challenge [28]. Recently, few-shot and zero-shot paradigms have emerged. Few-shot methods compare query images to a handful of normal support samples—either with strong vision encoders alone (e.g., AnomalyDINO [29], VisionAD [30]) or with VLMs. Zero-shot methods leverage VLMs by computing similarities to prompts describing normality versus abnormality, or by learning anomaly-aware prompts/tokens without any target-domain training, enabling broader out-of-the-box transfer (e.g., WinCLIP [31], AnomalyCLIP [32], AdaCLIP [33], VCP-CLIP [34]).

**Video Anomaly Detection (VAD)** refers to the process of automatically identifying unusual or abnormal events in video data that deviate from expected patterns of behavior. It has become an essential technology in a wide range of real-world applications, including traffic monitoring [35] and intelligent surveillance [36]. Within the domain of industrial manufacturing, anomalies come in various forms, such as assembling components in an incorrect sequence, exceeding the expected time constraints while performing tasks, etc. Being able to automatically detect diverse and often unpredictable anomalous events from video streams becomes difficult. The heterogeneity of these anomalies poses challenges in developing a general framework that addresses all possible scenarios.

**Time-Series Anomaly Detection (TSAD)** refers to the task of identifying irregular patterns or deviations in sequential data. In manufacturing environments, time series data is one of the most common types of data generated by modern factories. Anomalies often appear as point anomalies (individual extremes), contextual anomalies, or collective/subsequence anomalies [37]. In factories, such anomalies may show up as abnormal cycle times, sensor reading drifts, or emergent deviations across multiple channels. These anomalies can be early indicators of equipment failure, process inefficiencies, or safety hazards, making their timely detection critical for minimizing downtime and ensuring operational reliability.

TABLE I: Common Datasets for Prototype Design

| Dataset                                 | Year | Type      | #Samples    | Missions                                                            |
|-----------------------------------------|------|-----------|-------------|---------------------------------------------------------------------|
| ABC-Dataset [42]                        | 2019 | CAD Model | 1 million   | Patch segmentation/Geometric feature detection/Shape reconstruction |
| Scan-CAD Object Similarity Dataset [43] | 2019 | CAD Model | 14,225      | Object Retrieval                                                    |
| SketchGraphs [44]                       | 2020 | Sketches  | 15 million  | Sketches Design for Parametric CAD                                  |
| DeepCAD [9]                             | 2021 | CAD Model | 178,238     | Common Sequence for CAD Design                                      |
| Fusion 360 Gallery Dataset [45]–[47]    | 2021 | CAD Model | ~20k        | Object assembly, reconstruction, and segmentation                   |
| FloorPlanCAD Dataset [48]               | 2021 | CAD Model | 15,663      | Vector-grained and symbol-grained CAD Research                      |
| Automatic 3D CAD Models [49]            | 2023 | CAD Model | 2,981       | CAD reconstruction                                                  |
| CAD-Estate [50]                         | 2023 | CAD Model | 101k        | CAD reconstruction, 3D object alignment, and pose estimation        |
| BIKED++ [51]                            | 2024 | CAD Model | 1.4 million | Parametric CAD design for bikes                                     |

Foundation models tailored for TSAD should offer a scalable and generalizable approach to detect anomalies across diverse industrial scenarios. Traditional methods are often categorized into distance-based [38], density-based [39], and prediction-based [40] [41] families, operating through pipelines of sliding-window preprocessing, detection, score computation, and post-processing. However, many of these methods falter when cross-domain generalization or rare anomaly types are encountered. By leveraging large-scale pre-training on heterogeneous time series data, these models can capture complex temporal dependencies and subtle patterns that traditional rule-based or supervised methods often miss. Moreover, they enable rapid adaptation to new environments with minimal labeled data, which is especially valuable in industrial settings where annotation is costly and anomalies are rare. Integrating TSAD-capable foundation models into industrial AI systems enhances predictive maintenance, quality control, and energy optimization, ultimately driving smarter, more resilient manufacturing operations.

**Motion Detection** has long been a critical challenge in the field of robotics, primarily because researchers aim to enable robots to imitate human movements with high fidelity. However, in the field of industrial manufacturing, its value goes beyond imitation. Building a semantically meaningful motion vector space, such that similar motions are mapped to similar vector representations, can significantly enhance the understanding, classification, and optimization of manufacturing processes. Such a model can serve the following purposes:

- **Monitoring and compliance:** Ensure that every step of the process adheres to standard operating procedures (SOPs) through continuous surveillance.
- **Real-time anomaly detection:** Identifying deviations promptly to prevent defective products, reduce scrap, and minimize costly rework.
- **Security and safety assurance:** Detecting unusual or unauthorized events that should not occur within the manufacturing environment.
- **Process optimization:** Refine and improve SOPs for greater efficiency through post-process analysis of detected actions and anomalies.

To manage these challenges, a foundation model capable of detecting and categorizing a wide range of actions is essential.

## IV. DATASETS AND EVALUATION METRICS

### A. Design Datasets

Several publicly available CAD datasets are serving for design, as shown in Table I. Some datasets have inheritance relationships. For example, the Automatic 3D CAD Models dataset is collected from Fusion 360 gallery and several testing cases are collected from the ABC dataset.

ABC-Dataset [42] represents one of the extensive repositories of CAD data, encompassing a vast collection of CAD models, curves, and patches. Specifically designed to support geometric deep learning beyond simple classification, it facilitates advanced downstream tasks such as patch segmentation, geometric feature detection, and shape reconstruction. The dataset comprises one million CAD models provided in versatile formats, including .step, .stl, and .obj, ensuring broad compatibility for diverse downstream tasks. The Scan-CAD Object Similarity Dataset [43] is specifically curated for cross-modal object retrieval tasks. It provides paired correspondences between scanned 3D models (representing real-world sensor data) and their ideal CAD geometries. This dataset is instrumental for investigating the joint embedding space of CAD models, enabling researchers to bridge the gap between noisy scans and idealized designs. SketchGraphs [44] and DeepCAD [9] approach CAD design through parametric sketches. By establishing standardized CAD descriptions at the sketch level, these datasets enable researchers to model designs with high granularity, where complex geometries are constructed from fundamental geometric primitives and constraints. Extending from 2D sequential sketches, the Fusion 360 Gallery Dataset series [45]–[47] provides a comprehensive framework for investigating 3D CAD design through the lens of programmatic construction. These datasets facilitate research across diverse challenges, including automated assembly, high-fidelity reconstruction, and semantic segmentation, bridging the gap between sequential design operations and final geometric outputs. The approach proposed by Zhang et al. [49] focuses on the automated reconstruction of fine-grained CAD models from a limited set of orthographic drawings. By synthesizing 3D geometries from 2D projections, this model facilitates the completion of CAD designs even when only partial or sparse information is available, streamlining the transition from traditional drafting to digital modeling. The FloorPlanCAD Dataset [48] and BIKED++ [51] extend the scope of CAD research from isolated geomet-

ric primitives to complex, system-level objects. FloorPlan-CAD focuses on large-scale vectorized real estate layouts, while BIKED++ provides a multimodal repository of over 1.4 million bicycle designs. Together, these datasets enable IFMs to model intricate parametric relationships and structural inter-dependencies within specialized engineering domains. Furthermore, by leveraging real-world imagery and video sequences [50], 6DoF object alignment and pose estimation enable designers to utilize existing physical products as a baseline. This methodology facilitates the integration of real-world assets into the CAD workflow, significantly reducing the manual effort required for designing from scratch and streamlining the reverse engineering process.

### B. Quality Control Datasets

Quality control is a central application for IFMs, including defect detection, logical validation, and motion-level monitoring. Existing datasets can be broadly organized according to the type of anomaly or signal they emphasize, ranging from appearance-based defects to relational inconsistencies and temporal deviations in production activities.

**Visual Anomaly** focuses on appearance-level anomaly detection, where defects manifest as local visual irregularities on products or materials. Representative benchmarks include MVTec-AD [52], MVTec-3D [53], VisA [54], BTAD [55], MPDD [56], KSDD [57], Real-IAD [58], MANTA [59], and 3CAD [60]. These datasets span settings from single-class inspection to large-scale multi-category scenarios, and most adopt the standard unsupervised AD protocol, where training data contain only defect-free samples and anomalous instances are introduced only during inference. To support fine-grained localization, many datasets provide pixel-level ground truth. Recent benchmarks have also expanded sensing modalities and viewpoint diversity; for instance, MVTec-3D incorporates 3D point-cloud data, while Real-IAD and MANTA provide multi-view perspectives. Collectively, these benchmarks define the canonical evaluation landscape for industrial AD, emphasizing robustness and cross-category generalization.

**Logical Anomaly** identifies violations of contextual, relational, or functional constraints within real-world manufacturing. MVTec-LOCO [61] and Kaputt [62] provide benchmarks for logical AD. MVTec-LOCO introduces anomalies that necessitate reasoning about object co-occurrence and spatial configurations—even when individual components appear normal. The Kaputt benchmark further extends this paradigm by focusing on semantic and procedural inconsistencies, targeting errors that are visually subtle but invalid under specific task logic or process semantics. These datasets challenge models to infer and validate relational structures. This capability is distinct from, yet synergistic with, appearance-based detection, making it relevant for FMs that aim to bridge low-level perception with higher-level logical reasoning.

**Motion Detection and Process Segmentation** are critical for analyzing motion patterns and temporal process execution in industrial quality control. This requirement has motivated the development of datasets dedicated to video-based seg-

mentation and activity recognition. Early benchmarks such as Breakfast [63], 50 Salads [64], and YouTube Instructions [65] focus on long-horizon instructional videos. These works focus on temporal action segmentation, where the objective is to decompose complex human activities into a sequence of ordered steps. The MEKA dataset [66] advances this field by reorganizing single-task videos into multi-task sequences, facilitating the evaluation of compositional reasoning and cross-task segmentation. Nevertheless, benchmarks explicitly designed for industrial production environments remain scarce—a deficit largely attributed to the stringent privacy and security constraints inherent to manufacturing facilities. To bridge this gap, benchmarks such as Desktop Assembly [67], IPAD [68], IndustReal [69], and Assembly101 [70] provide factory-like recordings of multi-step assembly workflows. By combining these with instructional video insights, researchers can develop motion-aware FMs that bridge the gap between human-centric instructions and industrial operations.

### C. Evaluation Metrics

**Design Metrics:** In the context of concept design, expert evaluation serves as the primary benchmark for assessing generative outputs. Current methodologies typically employ a hybrid approach: human-centric assessments via crowd-sourced studies [71] and automated benchmarking leveraging LLMs [11] through Absolute Category Rating (ACR) protocols. By synthesizing multidimensional criteria—specifically feasibility, novelty, and utility—this framework ensures a robust validation of the functional and creative quality of generated CAD models.

Evaluating the performance of generative models in engineering design requires a multi-faceted approach that balances geometric fidelity with functional viability. Regenwetter et al. [72] established a comprehensive taxonomy for this purpose, categorizing evaluation into five dimensions: similarity, design exploration, constraint satisfaction, design quality, and conditioning. While these high-level categories define *what* to evaluate, the *how* depends on the availability of a reference design or ground-truth. In the context of prototype design, where ground-truth data is accessible, several quantitative metrics are employed to measure the alignment between generated outputs and design targets:

- **Geometric Fidelity Metrics:** To assess the point-to-point or primitive-to-primitive distance, *Chamfer Distance* [11], [15] and *Hausdorff Distance* are widely used, with the latter focusing on the maximum outlier error. Additionally, *Intersection over Union* (IoU) measures the volumetric overlap ratio, providing a robust indication of global shape accuracy.
- **Statistical and Probabilistic Metrics:** The *Log-likelihood* [13] is utilized to evaluate the likelihood of the generated samples within the learned CAD distribution. Other statistical measures, such as *Mean Squared Error* and *Mean Minimum Deviation*, quantify the average reconstruction error and distribution coverage, respectively.

- **Sequence and Structural Quality:** For models that generate CAD as a sequence of operations, *Accuracy*, *Precision*, *Recall*, and *F1-score* [15] are applied to evaluate the correctness of the primitive sequence compared to the ground-truth modeling steps.
- **Functional and Comparative Metrics:** Beyond geometric similarity, *Invalidity Ratio* assesses the model’s failure rate in producing manifold or executable CAD files, while *LZMA comparison* offers a proxy for structural complexity through compression-based similarity.

Despite this array of metrics, no single indicator has yet emerged that can comprehensively capture the holistic quality of industrial generation. Current research continues to navigate the trade-off between geometric precision (similarity) and the creative fulfillment of domain-specific functional requirements (design quality).

**Quality Control Metrics:** Evaluating the performance of FMs for AD necessitates a robust evaluation taxonomy that aligns with the multifaceted requirements of industrial environments. These include discriminative accuracy for outlier identification, spatial localization for defect pinpointing, temporal coherence for dynamic monitoring, and operational utility for practical deployment. To capture these dimensions, recent benchmarks have transitioned toward a multi-metric approach.

- **General-purpose AD metrics:** Across image, video, and time-series domains, the Area Under the Receiver Operating Characteristic Curve (AUROC) remains the standard for ranking anomalies, though it does not directly measure localization quality. To address the class imbalance inherent in industrial settings, Average Precision (AP)—or the area under the Precision–Recall curve—is frequently used to emphasize performance on the rare anomalous class. For deployment scenarios requiring a fixed decision threshold, metrics like the F1 score (and its best-achievable variant  $F1_{\max}$ ) are reported to evaluate the practical balance between false alarms and missed detections.
- **Spatial localization and segmentation metrics:** For spatial accuracy in images and videos, Intersection over Union (IoU) (or the Jaccard index) directly measures the overlap between predicted and ground-truth anomalies. However, as IoU is threshold-sensitive, industrial benchmarks prioritize region-aware metrics. Area Under the Per-Region Overlap curve (AUPRO) integrates overlap quality across varying false-positive rates (FPR), typically focusing on low-FPR regimes. Complementarily, the Area Under the Per-Image Overlap (AUPIMO) evaluates the detected anomalous area on a per-image basis, capturing subtle performance variances often obscured by AUPRO.
- **Temporal and event-level evaluation:** In time-series and video domains where anomalies manifest as contiguous events, frame-wise accuracy often proves misleading. Point-adjusted F1 marks an entire anomaly event as detected if any point within its interval is identified, though it can overestimate performance for long-duration events. Segment-wise F1 mitigates this by computing true and false positives

at the event level based on temporal overlap. For safety-critical or surveillance applications, Latency-aware Average Precision (LaAP) further incorporates detection timeliness by rewarding earlier identification. Beyond event detection, temporal alignment across sequences is captured by Mean over Frames (MoF), providing average frame-wise accuracy across all videos. For procedural tasks where action order is paramount, Edit Score [73] uses Levenshtein distance to measure the minimal changes needed to align predicted and ground-truth sequences, penalizing misarranged actions. Dynamic Time Warping (DTW) [74] complements this by allowing non-linear alignment, capturing temporal consistency without over-penalizing variations in action duration.

Modern industrial AD protocols leverage a diverse suite of ranking, localization, and temporal metrics. This multi-dimensional approach is critical for assessing models as it evaluates not only detection success but also the precision, consistency, and latency required for practical industrial deployment.

## V. CHALLENGES AND FUTURE DIRECTIONS

Despite rapid progress, deploying IFMs in real-world manufacturing environments remains constrained by several unresolved challenges, particularly in datasets and evaluation methodologies. Addressing these limitations is essential for transitioning IFMs from research prototypes to reliable industrial solutions.

From a data perspective, existing datasets often suffer from limited diversity, scale, and realism. Many benchmarks are collected in controlled laboratory settings and fail to capture the full variability of factory-floor conditions, such as sensor noise, domain shifts, and illumination conditions. Accurate object alignment is essential for minimizing background noise across different industrial settings; yet, jointly optimizing multiple ML modules remains an area of ongoing research.

Current evaluation protocols rely on task-specific, fragmented metrics that often lack a unified framework. Many benchmarks depend on simplified assumptions—such as static thresholds, frame-wise independence, and idealized anomaly definitions—which frequently result in over-optimistic performance results. Moreover, existing frameworks rarely account for critical practical factors, including false-alarm costs, detection latency, and the nuances of human-in-the-loop interaction. Consequently, there is an urgent need for more holistic, deployment-oriented metrics that bridge the gap between algorithmic performance and operational utility in complex industrial environments.

Advancing IFMs requires a more cohesive integration of multimodal data, architectural innovation, rigorous data curation, and robust evaluation design. By addressing these dataset- and metric-centric challenges, the research community can move toward industrial AI systems that are not only high-performing, but also trustworthy, scalable, and resilient to the complexities of modern manufacturing.



## VI. CONCLUSION

This overview paper has examined the recent advancements in Industrial Foundation Models (IFMs) for manufacturing, specifically detailing core architectures, applications in design and quality control, and the landscape of datasets and evaluation metrics. By identifying critical gaps in data curation and performance measurement, we highlight the requirements for successful practical deployment. We believe this work provides a strategic foundation for integrating IFMs into real-world manufacturing and serves as a roadmap for future research.

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